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HOMEWORK 2 Due March 3 at 11:59PM on Gradescope.

Note: In showing that function families below are not secure PRFs, exhaustive key search and birthday attacks are inefficient and will receive no credit. Your adversaries must be practical, making at most a handful of queries and performing only minor additional computation. For each adversary, clearly state its advantage and resource usage.

Problem 1. (30 points.) Define the family of functions $F\colon \{0,1\}^{128} \times \{0,1\}^{256} \to \{0,1\}^{128}$ by

$$F(K, M_1 \| M_2) = \mathsf{AES}(K, M_1) \oplus \mathsf{AES}(K, M_2)$$

for all $K \in \{0,1\}^{128}$ and $M_1, M_2 \in \{0,1\}^{128}$. Show that F is not a secure PRF.

Problem 2. (70 points.) Let $G\colon \{0,1\}^k \times \{0,1\}^\ell \to \{0,1\}^\ell$ be a family of functions (arbitrary but given, hence known to the adversary) and let $r \geq 1$ be an integer. The *r-round Feistel cipher associated to G* is the family of functions $G^{(r)}\colon \{0,1\}^k \times \{0,1\}^{2\ell} \to \{0,1\}^{2\ell}$ defined as follows for any key $K \in \{0,1\}^k$ and input $x \in \{0,1\}^{2\ell}$:

Algorithm $G^{(r)}(K, x)$:
  Parse x as $L_0 \| R_0$ where $|L_0| = |R_0| = \ell$
  For $i = 1$ to r do:
    $L_i \leftarrow R_{i-1}$; $R_i \leftarrow G(K, R_{i-1}) \oplus L_{i-1}$
  Return $L_r \| R_r$

(Part A — 20 points.) Show that $G^{(1)}$ is not a secure PRF.

(Part B — 30 points.) Show that $G^{(2)}$ is not a secure PRF.

(Part C — 20 points.) Now consider a two-key variant of the 2-round Feistel cipher. Specifically, define $H\colon \{0,1\}^{2k} \times \{0,1\}^{2\ell} \to \{0,1\}^{2\ell}$ by using independent round keys $K_1, K_2 \in \{0,1\}^k$:

Algorithm $H(K_1 \| K_2, x)$:
  Parse x as $L_0 \| R_0$ where $|L_0| = |R_0| = \ell$
  $L_1 \leftarrow R_0$; $R_1 \leftarrow G(K_1, R_0) \oplus L_0$
  $L_2 \leftarrow R_1$; $R_2 \leftarrow G(K_2, R_1) \oplus L_1$
  Return $L_2 \| R_2$

Show that H is not a secure PRF.

Optional Bonus Problem. The *ideal cipher model*, first proposed by Shannon, models a blockcipher as a family of *independent* random permutations, one per key. Concretely, for key-length k and block-length ℓ, an ideal cipher $\mathsf{IC}\colon \{0,1\}^k \times \{0,1\}^\ell \to \{0,1\}^\ell$ is one where for each fixed key $K \in \{0,1\}^k$, the function $\mathsf{IC}(K, \cdot)$ is an independently and uniformly random permutation on $\{0,1\}^\ell$, unknown to everyone until queried. An adversary in the ideal cipher model may query $\mathsf{IC}(K', M')$ for any (K', M') of its choice; the model guarantees consistency (same query always returns the same answer) and that the mapping for each fixed key remains a valid permutation.

(Part A.) In your own words, explain what the ideal cipher model assumes about a blockcipher and compare it to modeling the blockcipher as a secure PRF. In particular: is a real blockcipher like AES an ideal cipher? Discuss briefly.

(Part B.)

Quiz 2. Let $F\colon \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ be a function family for some $n \in \mathbb{N}$. We say that F is a *dual PRF* if F is a PRF and the function family F_{dual} is a PRF, where $F_{\mathsf{dual}}(K, x) := F(x, K)$ for all $x, K \in \{0,1\}^n$. Show that AES is *not* a dual PRF.

$AES_{dual}(K, x) = AES(x, K)$

WTS AES_{dual} is not a PRF.

Adversary $\mathbb{A}^{F_n(x)\{AES(x,K),\ \$\}}$

$y \leftarrow F_n(0^{128})$

$y' \leftarrow F_n(1^{128})$

$x \leftarrow AES^{-1}(0^{128}, y)$

$x' \leftarrow AES^{-1}(1^{128}, y')$

Ret $(x = x')$

Resources:
2 F_n queries
+ $2 \cdot T_{AES} + O(1)$

$Adv(\mathbb{A}) \stackrel{def}{=} Pr[REAL \Rightarrow 1] - Pr[RAND \Rightarrow 1]$

Claim 1. $Pr[REAL \Rightarrow 1] = 1$.

proof. In the REAL game, on line 1 $y = AES(0^{128}, K)$. On line 2 $y' = AES(1^{128}, K)$. on line 3 $x = K$. On line 4 $x' = K$

$\Rightarrow$ Return 1

Claim 2. In the RAND game, on line 1 $y = \$$. In line 2, $y' = \$'$

In line 3 $x = AES^{-1}(0^{128}, \$)$

In line 4 $x' = AES^{-1}(1^{128}, \$')$

$\rightarrow$ Since for any fixed K, $AES_K(\cdot)$ is a permutation, $Pr[x = x'] = 2^{-128}$.

$Pr[RAND \Rightarrow 1] = 2^{-128}$

(i) Write pseudocode for A^1_{eks} in the ideal cipher model.

(ii) Compute the TKR-advantage of A^1_{eks} as a closed-form expression in k and ℓ. **Hint:** The target key K is always consistent. Each other candidate key $K' \neq K$ is independently consistent with probability $1/2^\ell$. What is the probability that K is the *first* consistent key found?